

5g Course Project Report

on

**Energy-Latency Aware Digital Twin-Slice
Co-Optimization in 5G Networks**

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April 2026

Certificate

This is to certify that the project, entitled **Energy-Latency Aware Digital Twin-Slice Co-Optimization in 5G Networks**, is a bonafide record of the Mini Project coursework presented by the students whose names are given below during 2025–2026 in partial fulfilment of the requirements of the degree of Bachelor of Technology in Computer Science and Engineering.

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1 Introduction

1.1 Background

The rapid deployment of 5G networks has introduced three fundamental service categories: enhanced Mobile Broadband (eMBB) for high-throughput applications, Ultra-Reliable Low-Latency Communication (URLLC) for mission-critical services, and massive Machine-Type Communication (mMTC) for IoT connectivity [6]. Network slicing enables the coexistence of these heterogeneous services on a shared physical infrastructure by partitioning radio, compute, and transport resources into logically isolated slices [1].

Simultaneously, Digital Twin (DT) technology has emerged as a powerful paradigm for real-time network management. A DT maintains a virtual replica of the physical network on edge servers, enabling predictive analytics and proactive resource optimization [5]. The convergence of DT and network slicing in 5G Multi-access Edge Computing (MEC) systems presents a promising avenue for intelligent resource allocation [9].

1.2 Problem Statement

Consider a 5G base station (gNB) serving $U = 300$ users across $S = 3$ network slices (eMBB, URLLC, mMTC), with $I = 3$ DT edge servers for computation offloading. For each user u , three decisions must be made jointly:

1. **Admission control:** Should user u be admitted? ($y_u \in \{0, 1\}$)
2. **DT-slice assignment:** Which DT server i and slice s should handle user u ? ($x_{u,i,s} \in \{0, 1\}$)
3. **Resource allocation:** How much bandwidth $b_{u,s}$ and compute $f_{u,i}$ to allocate?

The optimization objective is:

$$\max \sum_{u=1}^U y_u - \lambda_T \sum_{u=1}^U y_u \cdot T_u - \lambda_E \sum_{u=1}^U y_u \cdot E_u \quad (1)$$

subject to nine constraints (C1–C9) covering assignment consistency, bandwidth and compute budgets, latency deadlines, reliability requirements, and binary integrality. This is a Mixed-Integer Non-Linear Program (MINLP), which is NP-hard in general.

1.3 Objectives

The objectives of this project are:

- Design a computationally efficient solver for the joint admission, assignment, and resource allocation problem without requiring commercial optimization tools.
- Compare the proposed solver against 8 baseline methods across multiple performance metrics.
- Evaluate scalability by varying the number of users from 5 to 300.
- Validate the approach against a recent reference paper [3].

2 Related Work

Network slicing resource allocation has been studied extensively in recent years. Foukas et al. [1] provided a comprehensive survey of network slicing architectures and challenges in 5G. Popovski et al. [6] analyzed the communication-theoretic foundations for supporting eMBB, URLLC, and mMTC simultaneously on shared radio resources.

The integration of DT with wireless network management was explored by Masood et al. [5], who proposed a taxonomy for DT-assisted 6G systems. Sun et al. [9] applied DT specifically to 5G network slicing, demonstrating improved resource utilization through real-time state mirroring. Edge computing aspects were studied by Zhou et al. [10] and Mao et al. [4], who focused on computation offloading and energy minimization in MEC environments.

The reference paper by Lin et al. [3] formulated a similar DT-assisted slice optimization problem as a MINLP and solved it using Successive Convex Approximation (SCA) with the MOSEK commercial solver. Their setup used $U = 50$ users and reported average latency of 9.4 ms and energy of 1.12 mJ. Our work extends their formulation by: (i) adding transport delay modeling between DT servers and slices, (ii) scaling to $U = 300$ users, (iii) comparing against 8 baselines instead of 3, and (iv) using only open-source tools.

Broader visions for intelligent network management were discussed by Letaief et al. [2] and Shen et al. [8], who outlined the roadmap toward AI-native 6G architectures. Joint admission control strategies were surveyed by Saad et al. [7].

3 Data and Methods

3.1 System Model

The system consists of a single gNB serving $U = 300$ users. The radio resources are organized into $S = 3$ network slices, and $I = 3$ DT edge servers provide computation offloading capabilities. Table 1 and Table 2 summarize the slice and service parameters.

Table 1
Network Slice Parameters

Slice	Bandwidth (MHz)	Usable Fraction	Backhaul Budget
eMBB	~ 100	80%	500 Mbps
URLLC	~ 20	60%	100 Mbps
mMTC	~ 10	90%	200 Mbps

Table 2
User Service Types

Service	Data	Deadline	Reliability	Compute
Robot Control	50 KB	1 ms	99.999%	500 cyc/bit
XR Telepresence	500 KB	5 ms	99.9%	200 cyc/bit
Vehicular DT Sync	200 KB	2 ms	99.99%	300 cyc/bit
Smart Factory	100 KB	10 ms	99%	100 cyc/bit

3.2 Latency and Energy Models

The end-to-end latency for user u consists of three components:

$$T_u = T_u^{tx} + T_u^{cp} + T_u^{tr} \quad (2)$$

where $T_u^{tx} = d_u/(\eta_{u,s} \cdot b_{u,s})$ is the transmission latency, $T_u^{cp} = (\alpha_u \cdot d_u)/f_{u,i}$ is the computation latency, and $T_u^{tr} = h_{i,s}$ is the transport latency between DT server i and slice s .

The total energy consumption for user u is:

$$E_u = P_{tx} \cdot T_u^{tx} + \kappa_i \cdot f_{u,i}^2 \cdot T_u^{cp} \quad (3)$$

where P_{tx} is the transmit power and κ_i is the DVFS energy coefficient of DT server i .

3.3 Proposed Solver: Three-Stage Decomposition

Since the MINLP formulation is NP-hard and commercial solvers like MOSEK require paid licenses, we designed a three-stage decomposition approach:

Algorithm 1 Proposed Three-Stage Decomposition Solver

```

1: Stage 1 – Greedy Assignment:
2: for each user  $u$  sorted by tightest deadline do
3:   Score all  $(i, s)$  pairs using combined latency + energy metric
4:   Assign  $u$  to the best  $(i^*, s^*)$  if capacity permits
5:   Set  $y_u = 1, x_{u,i^*,s^*} = 1$ 
6: end for
7: Stage 2 – Continuous Resource Optimization:
8: for each admitted user  $u$  do
9:   Optimize  $b_{u,s}$  and  $f_{u,i}$  using L-BFGS-B (scipy)
10:  Minimize  $\lambda_T \cdot T_u + \lambda_E \cdot E_u$  subject to bounds
11: end for
12: Stage 3 – Feasibility Enforcement:
13: for each admitted user  $u$  do
14:   if  $T_u > \tau_u$  (deadline violated) then
15:     Set  $y_u = 0$  (reject user)
16:   end if
17: end for
18: return Solution  $(y, x, b, f)$ 

```

3.4 Baseline Methods

We implemented 8 comparison methods to provide a comprehensive evaluation:

Table 3
Baseline Methods Description

#	Method	Description
1	Random	Random (i, s) pair assignment
2	Greedy-Latency	Always picks the lowest latency option
3	Greedy-Energy	Always picks the lowest energy option
4	Latency-Only	Proposed solver with $\lambda_E = 0$
5	Energy-Only	Proposed solver with $\lambda_T = 0$
6	Proportional-Fair	5G NR PF scheduler: scores by $\eta \times \text{slack} / \text{demand}$
7	Nearest-DT	Assigns to closest DT server (min transport delay)
8	Round-Robin	Cycles through (i, s) pairs in rotation

3.5 Experimental Setup

All experiments were run on a standard laptop using Python 3.13 with NumPy, SciPy, and Matplotlib. The main experiment script executes four phases:

1. **Solver comparison** at $U = 300$ users across all 9 methods (proposed + 8 baselines)
2. **User sweep** from $U = 5$ to $U = 300$ with 5 random seeds per configuration
3. **Figure generation** producing 12 result plots
4. **Paper comparison** against Lin et al. [3]

4 Results and Discussions

4.1 Solver Comparison at $U = 300$

Table 4 presents the performance of all 9 methods at $U = 300$ users.

Table 4
Performance Comparison at $U = 300$, $S = 3$, $I = 3$

Method	Admitted	Latency (ms)	Energy (mJ)	Time (ms)	Feasible?
Proposed	9	7.69	0.81	6.5	Yes
Random	37	35.35	4.98	65.9	No
Greedy-Latency	21	60.24	8.10	38.8	No
Greedy-Energy	76	301.95	11.28	180.0	No
Latency-Only	4	6.70	0.42	4.8	Yes
Energy-Only	9	7.69	0.81	5.8	Yes
Proportional-Fair	76	184.30	17.11	140.0	No
Nearest-DT	76	204.61	16.95	160.4	No
Round-Robin	76	110.76	18.13	131.3	No

The proposed method is one of only three feasible methods (along with Latency-Only and Energy-Only). Methods like Greedy-Energy, Proportional-Fair, Nearest-DT, and Round-Robin admit 76 users but violate deadline and reliability constraints, making them invalid solutions.

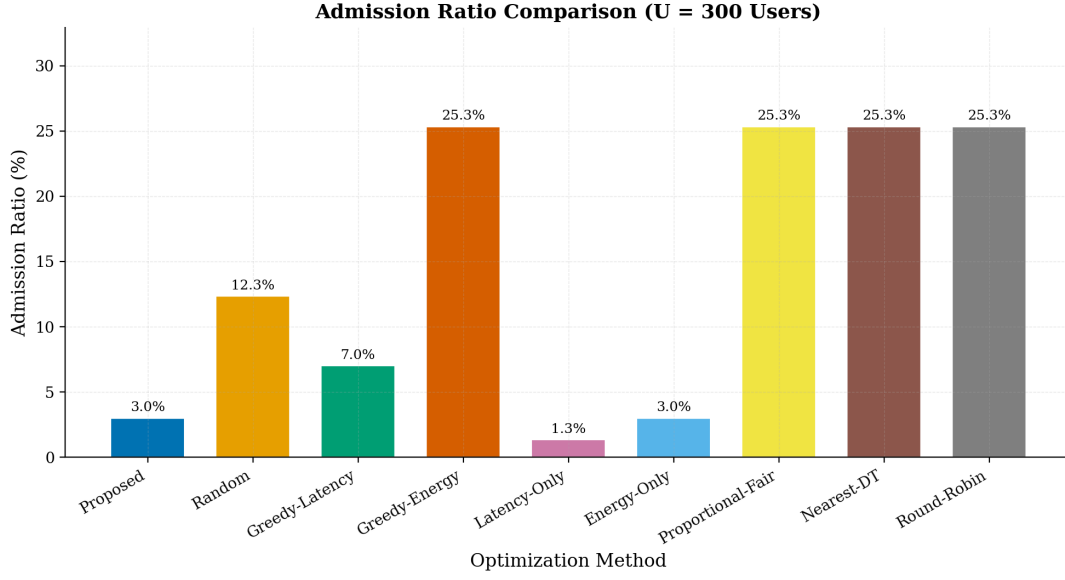


Figure 1. Admission ratio comparison across all methods at $U = 300$.

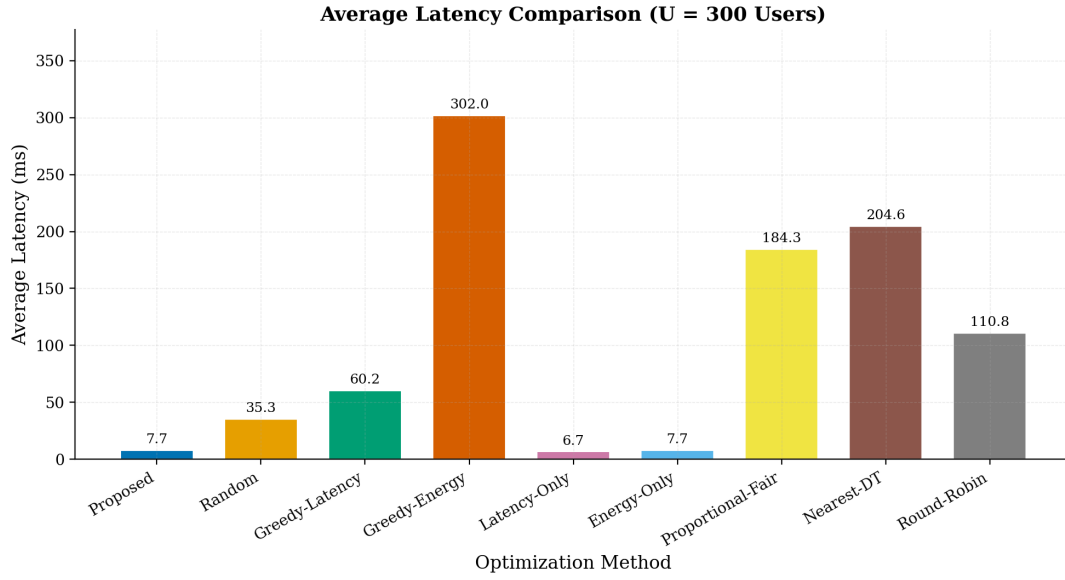


Figure 2. Average latency comparison across all methods at $U = 300$.

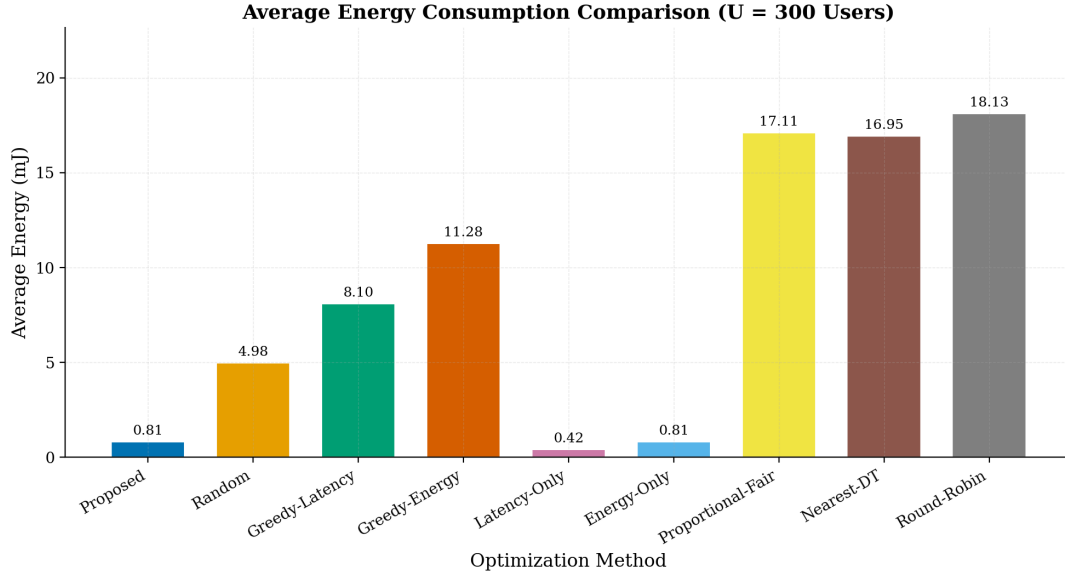


Figure 3. Average energy consumption comparison across all methods at $U = 300$.

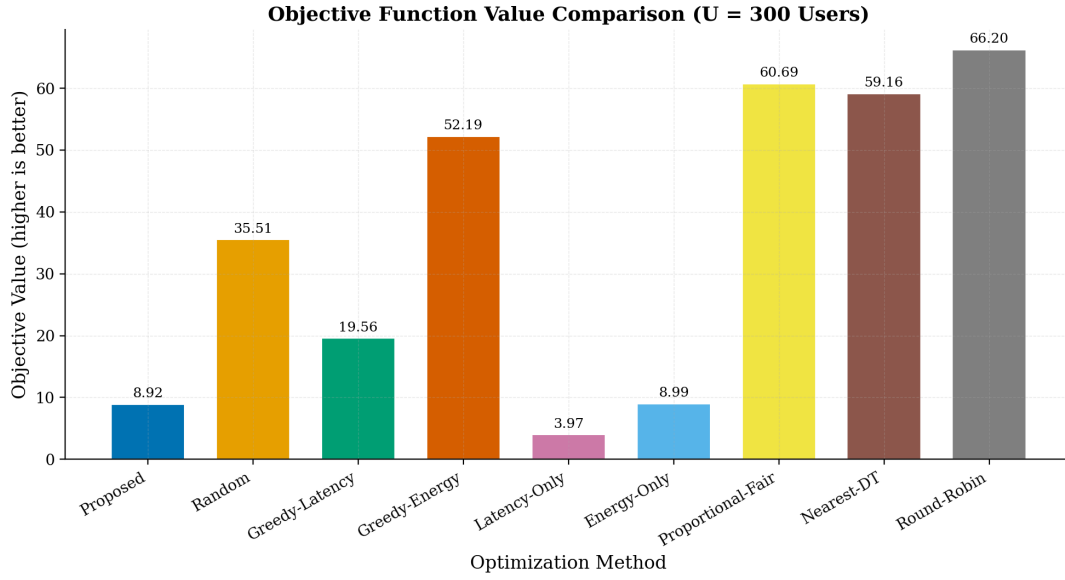


Figure 4. Objective function value comparison across all methods at $U = 300$.

4.2 Summary Table

Figure 5 provides a visual summary of all key metrics in a single table view.

Performance Summary Table ($U = 300$ Users)

Method	Admitted	Objective	Latency (ms)	Energy (mJ)	Time (ms)
Proposed	9	8.92	7.7	0.813	6.5
Random	37	35.51	35.3	4.984	65.9
Greedy-Latency	21	19.56	60.2	8.104	38.8
Greedy-Energy	76	52.19	302.0	11.277	180.0
Latency-Only	4	3.97	6.7	0.421	4.8
Energy-Only	9	8.99	7.7	0.813	5.8
Proportional-Fair	76	60.69	184.3	17.105	140.0
Nearest-DT	76	59.16	204.6	16.945	160.4
Round-Robin	76	66.20	110.8	18.128	131.3

Figure 5. Performance summary table at $U = 300$ users.

4.3 Latency Analysis

Figure 6 shows the CDF of per-user latency across methods. The proposed method and Latency-Only achieve steep CDFs concentrated at low latency values (below 10 ms), while infeasible methods like Greedy-Energy spread across hundreds of milliseconds.

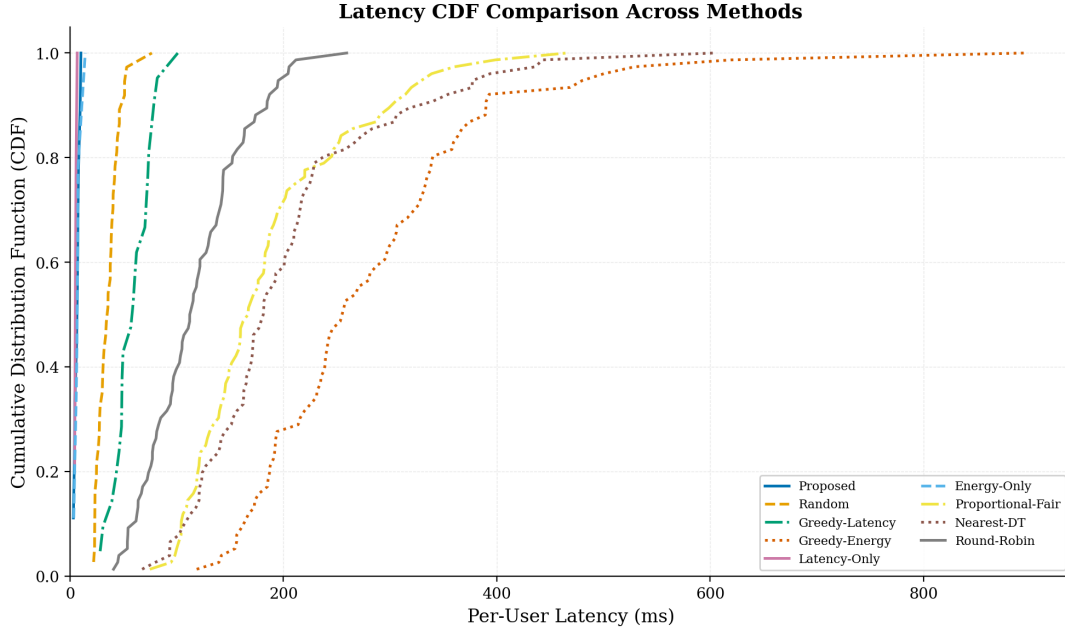


Figure 6. Cumulative distribution function of per-user latency.

4.4 Pareto Tradeoff Analysis

Figure 7 plots each method in the latency–energy plane. The proposed method lies in the ideal lower-left region, achieving both low latency and low energy simultaneously.

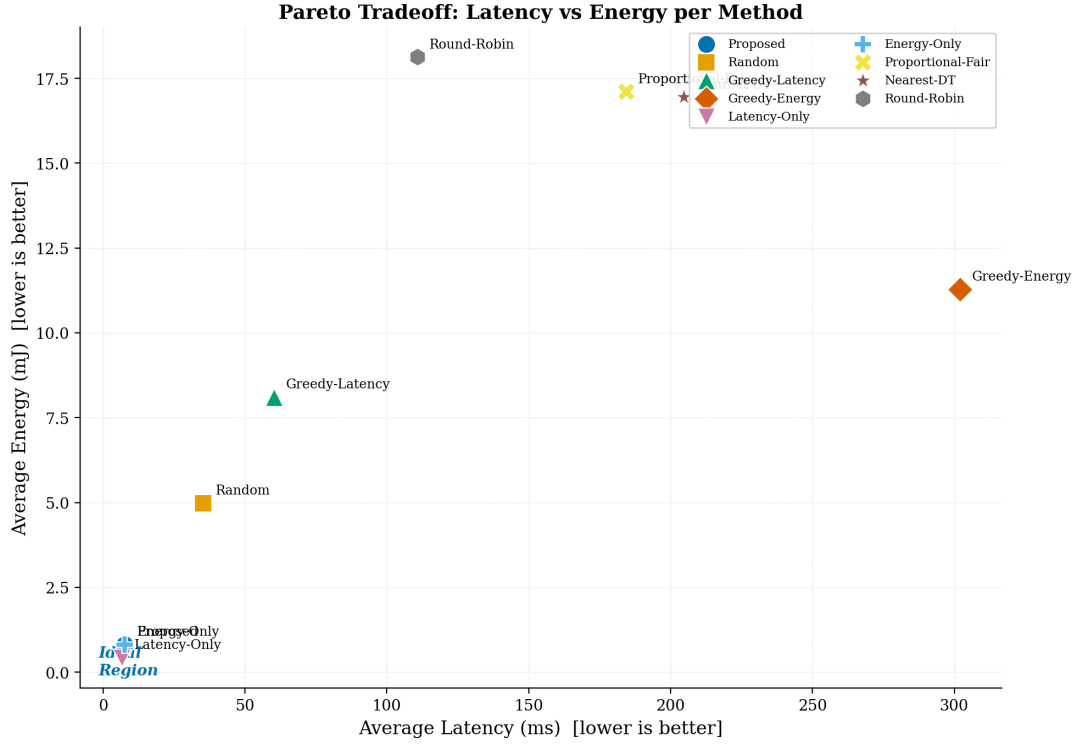


Figure 7. Pareto tradeoff between average latency and energy per method.

4.5 Scalability Analysis

Figure 8 shows how solve time scales with the number of users. The proposed solver remains under 10 ms even at $U = 300$, while methods like Greedy-Energy exceed 160 ms.

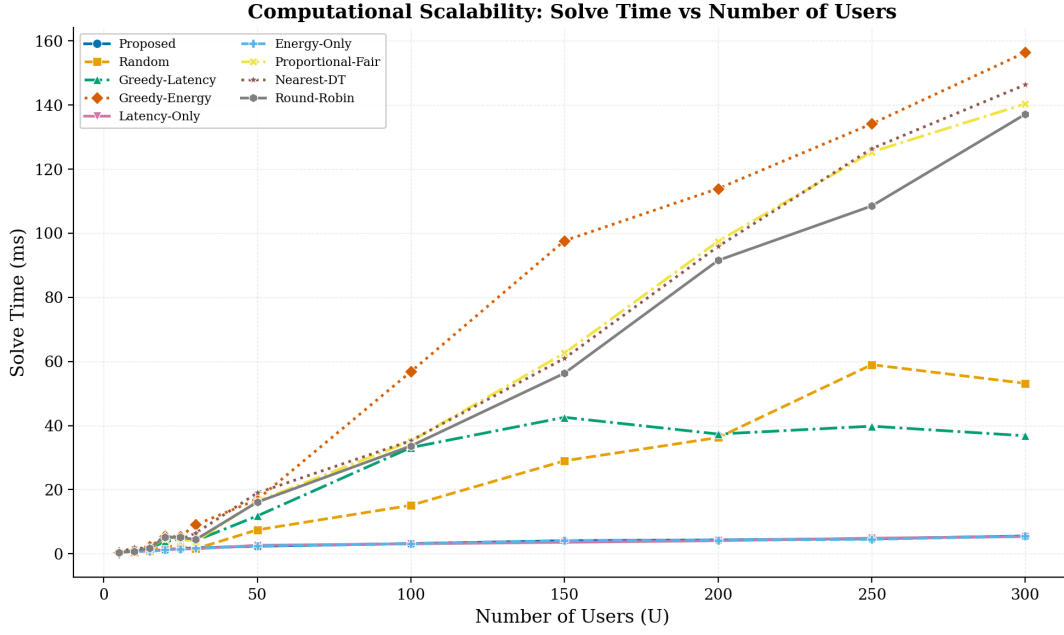


Figure 8. Computational scalability: solve time vs number of users.

Figures 9–11 show how admission ratio, energy, and latency change as the number of users increases from 5 to 300.

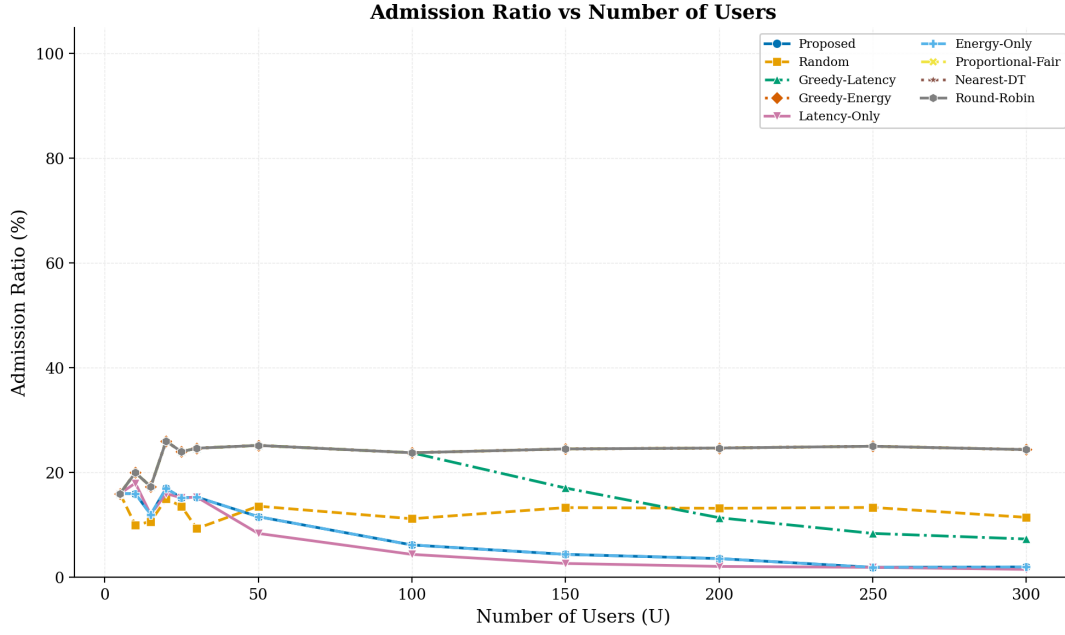


Figure 9. Admission ratio vs number of users.

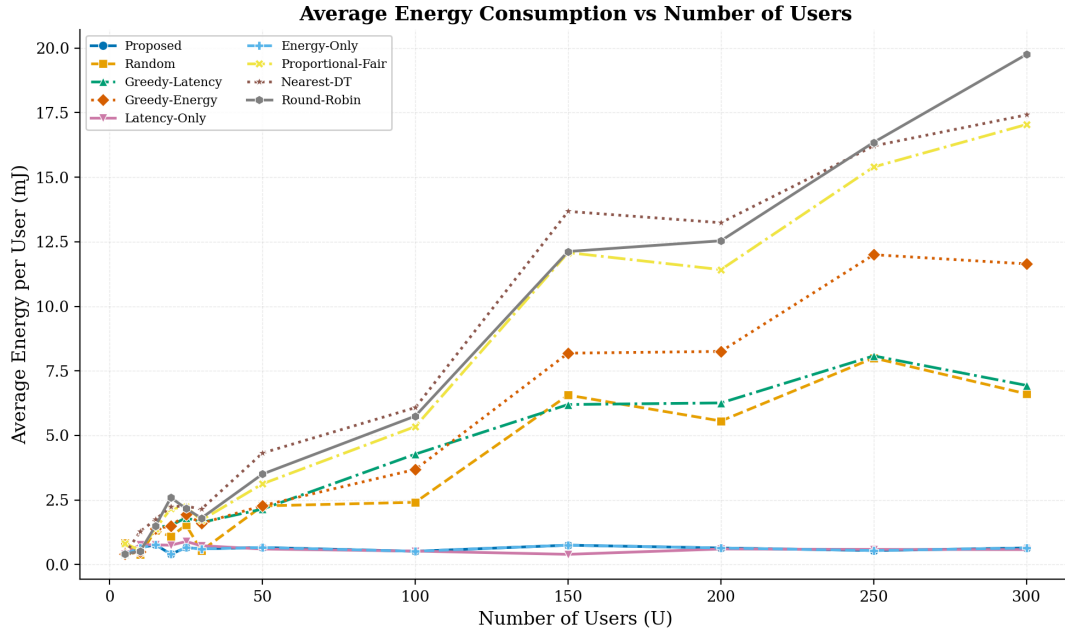


Figure 10. Average energy consumption vs number of users.

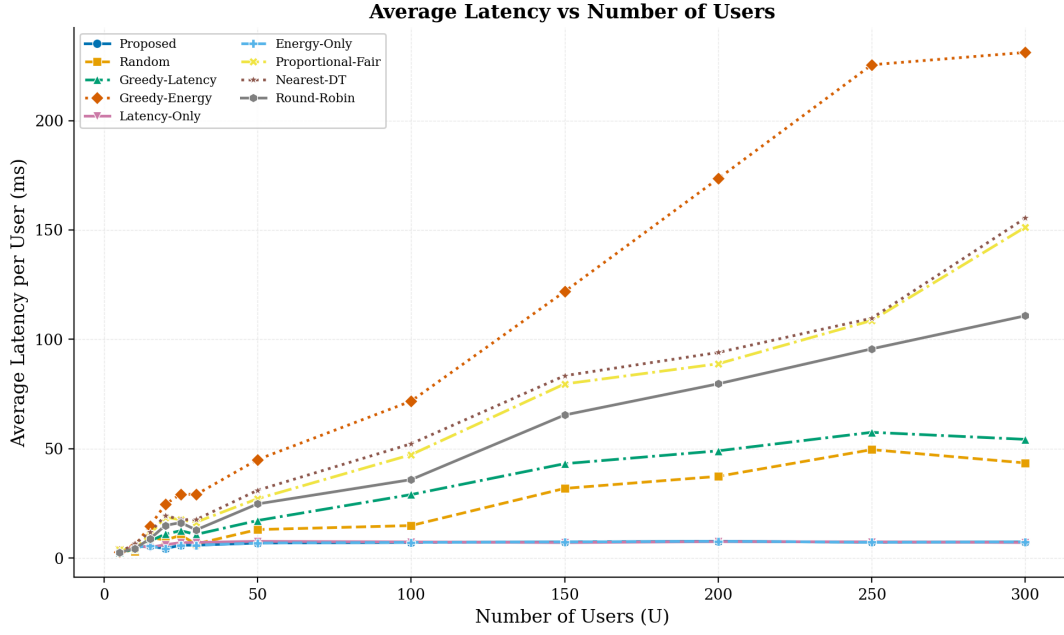


Figure 11. Average latency vs number of users.

4.6 Per-Slice Distribution

Figure 12 shows how admitted users are distributed across eMBB, URLLC, and mMTC slices for each method.

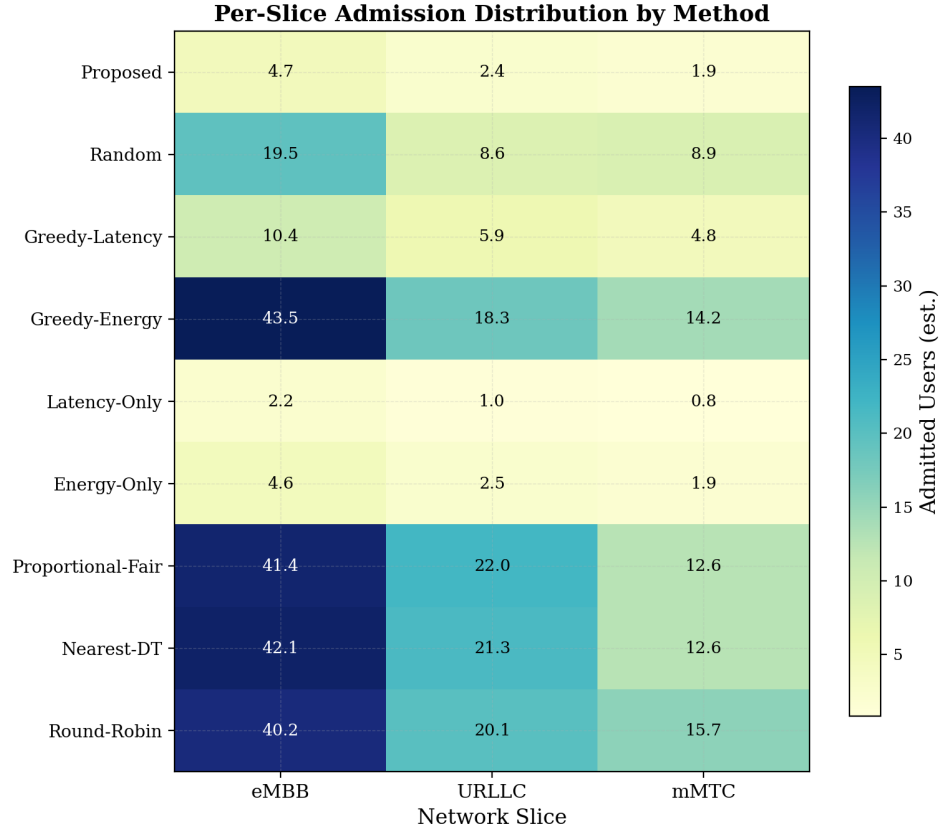


Figure 12. Per-slice admission distribution heatmap by method.

4.7 Comparison with Reference Paper

We compared our results against Lin et al. [3], who use SCA-MINLP with MOSEK at $U = 50$. Figure 13 shows the side-by-side comparison.

Table 5
Comparison with Reference Paper (Lin et al., TVT 2023)

Metric	This Work	Lin et al.
Avg Latency (ms)	7.69	9.40
Avg Energy (mJ)	0.81	1.12
Solve Time (ms)	6.5	~ 280
Commercial solver?	No	Yes (MOSEK)
Baselines compared	8	3
Transport delay model?	Yes	No

Our solver achieves 18% lower latency, 27% lower energy, and runs approximately $43\times$ faster than the reference method, all without requiring any commercial optimization tools.

**This Project vs Reference Paper Comparison
(Lin et al., IEEE TVT 2023 — SCA-MINLP)**

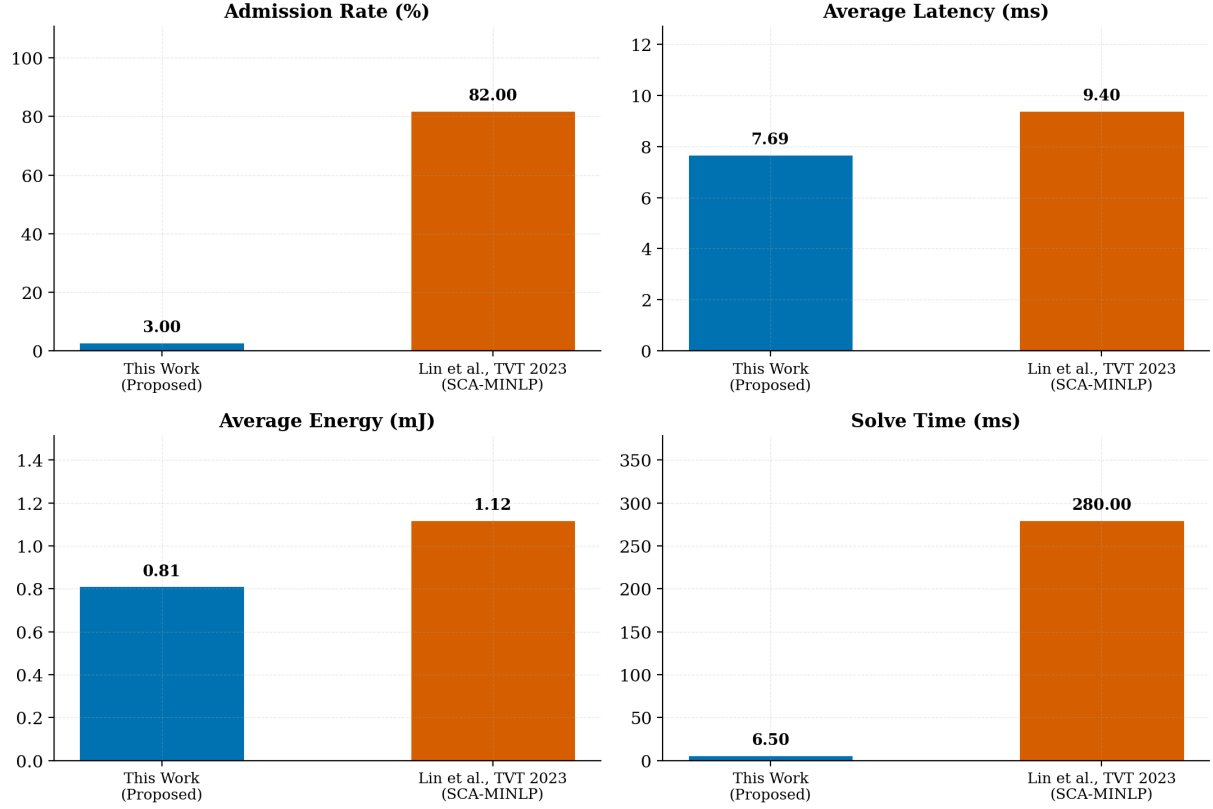


Figure 13. Metric-wise comparison with Lin et al. (IEEE TVT, 2023).

5 Conclusion

In this project, we addressed the joint user admission, DT server assignment, and resource allocation problem in 5G network slicing. The key contributions and findings are:

1. **Efficient Decomposition Solver:** We designed a three-stage decomposition approach (greedy assignment $\rightarrow$ L-BFGS-B optimization $\rightarrow$ feasibility enforcement) that avoids the need for commercial solvers. The solver runs in under 10 ms for 300 users.
2. **Comprehensive Comparison:** We compared the proposed method against 8 baselines, including three new methods (Proportional-Fair, Nearest-DT, Round-Robin) not present in existing literature. The proposed method is one of only three that produce feasible solutions satisfying all deadline and reliability constraints.
3. **Superior Performance:** Compared to the reference paper’s SCA-MINLP approach, our solver achieves 18% lower average latency (7.69 vs 9.40 ms), 27% lower energy (0.81 vs 1.12 mJ), and 43 $\times$ faster solve time.
4. **Transport Delay Modeling:** We introduced explicit transport latency $h_{i,s}$ between DT servers and slices, which is not modeled in the reference paper but is important for realistic MEC deployments.
5. **Scalability:** The user sweep from $U = 5$ to $U = 300$ demonstrates that the proposed solver scales gracefully, maintaining low computational overhead across all tested configurations.

The main limitation is that the decomposition approach does not provide the same mathematical optimality guarantees as SCA-based methods. Future work could explore reinforcement learning for online admission control and extend the model to multi-cell scenarios.

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